

Charles Karpati

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Summary of Experience

Accomplished software developer with over 10 years of experience in creating innovative solutions and leading technical teams. Expertise in full-stack development, computer vision, and machine learning, with a strong focus on enhancing user experiences through technology. Proven track record in automating complex processes and driving business value through technical decision-making. Proficient in Node.js, Python, and advanced rendering techniques, with a solid background in GIS technologies and cloud solutions.

Work History

Founder

March 2024 - Current

FeelsGoodGoods

- Introduced 'Sign-In' features for two different products, reducing bounce rates by 20% using passport.js, metamask, google OAuth2, JWT tokens, and sessions.
- Added 'Pay with Crypto' and 'Pay with Stripe' functionality to sites, allowing them to take payment on-page and save approximately 3% per transaction.
- Implemented a Telegram chat recap service using AI, servicing six communities across three cities, nearly 1,000 members, utilizing Cron, Node.js, Python, and PHP for automated updates.
- Developed an AI-powered Chrome extension designed to automate information retrieval and assist in document generation and online responses, employing technologies like Webpack, React, and SQLite.
- Engineered a fully operational, self-hosted private cloud using home lab infrastructure, implementing secure and reliable data management practices.
- Created a comprehensive continuous integration and delivery (CI/CD) and DevOps pipelines to automate and streamline deployment processes, leveraging Docker, SFTP, and SSH.

Chief Technology Officer

May 2023 - February 2024

Addy-AI

- Co-founded and served as Chief Technology Officer for AddyAI, an automated AI email response service with over 10,000 users, demonstrating a blend of entrepreneurial vision and technical leadership.
- Played a key role in achieving a \$10M valuation for AddyAI during pre-seed funding, engaging in successful pitches to Fortune 500 companies and venture capitalists.
- Secured \$400k in venture capital support, leveraging relationships with programs like Google for Startups to fuel company growth.
- Created Langdrive, an open-source tool for automated LLM fine-tuning, simplifying a previously highly technical task into a no-code solution.
- Experience migrating on-premises applications to containerized solutions in AWS (EKS) and updating applications to use AWS native tools (RDS).
- Developed CLI and API tools for data preparation, model training, and cloud deployment to Hugging-Face or Heroku, simplifying LLM operations.
- Implemented CI/CD pipelines utilizing SFTP, SSH, and Docker Compose to automate and streamline the software deployment process.

Software Developer

March 2022 - May 2023

Voxels.com

- Enhanced the platform's Web3 capabilities and maintained relations with DAOs and partners at OpenSea and Infura.
- Introduced in-game crypto-transactions and Proof-of-Attendance (POAP) using Babylon.js and Ethers.js.
- Spearheaded efforts to improve the quality and performance of the platform's real-time world map feature, leveraging technologies like Leaflet and Redis.

- Worked on a monolithic codebase with microservices in TypeScript, promoting efficient and modular software development using services like Docker, Digital Ocean, Heroku, AWS, and GitHub Actions.

Lecturer

January 2020 - November 2021

National Science Foundation

- Instructed graduate-level bi-annual courses in cohorts of 5 on Python for Big Data, enhancing students' skills in data analysis and visualization through hands-on learning.
- Designed and developed course curriculum in partnership with universities, ensuring that the material met academic and professional standards.
- Mentored students on data science projects, providing guidance and feedback to help them deliver high-quality analyses and presentations.
- Created interactive lessons focused on data visualization, using real-world examples to help students effectively analyze and present complex data sets.

Web Developer

January 2017 - November 2021

Baltimore Neighborhood Indicators Association

- Developed interactive GIS maps and dashboards to communicate data to city-based stakeholders using tools like Webpack, React, and AWS.
- Automated generation of critical statistics used by the organization leveraging GeoPandas, Vega, and Plotly.
- Collaborated closely using Microsoft Teams and SharePoint for effective communication and resource sharing.
- Experience developing and working with open-source and commercial GIS technologies and data, including OGC GIS service standards (WFS, WMS), ESRI map services, ArcGIS JavaScript APIs, Leaflet, GDAL, and PostGIS.

Contract Developer

January 2009 - Present

- Designed and developed websites for diverse clients, including law firms, art schools, cinemas, and elder care facilities, using Jekyll, GitHub, and Google Analytics.
- Created 'Green Patterns Map,' a web-based visualization tool to track and promote water conservation efforts in Baltimore, in partnership with federal and local agencies, using React, PHP, and PostgreSQL.
- Built the 'NEAR Library', a virtual library in the metaverse, funded by a grant, using the Near Crypto API and Express.
- Collaborated with the Baltimore City Mayor's Office to develop the 'Equity Tracker,' a tool used by city agencies to monitor equity goals, employing Microsoft 365, Power BI, Power Automate, SharePoint, and Power Forms.

Education

MS in Information Systems

University of Maryland, Baltimore County

BS in Information Systems

University of Maryland, Baltimore County

Awards

GitHub Open-Source Accelerator Alumni (AddyAI, 2024)

Venture Bridge Alumni, Carnegie Mellon Accelerator (AddyAI, 2023)

Tools

HTML, CSS, JavaScript, TypeScript, Node.js, Python, SQL, PostgreSQL, Redis, Bash, PHP, Java, React, Docker, AWS, Firebase, OAuth2, WordPress, Git, GitHub, Azure, Slack, Figma, Leaflet, Three.js, Pandoc, TensorFlow, OpenAI, HuggingFace, Pandas, Solidity, Power BI, Tableau, OGC GIS service standards (WFS, WMS), ESRI map services, ArcGIS JavaScript APIs, GDAL, PostGIS, RDS, EKS